Year 2 Quantum Mechanics - Example Sheet 4

All of the material for this sheet has already been covered.

1. The nucleus of ^{232}Th emits α-particles of energy E = 4.05 MeV and has a half-life of 1.39×10^{10} years. The nucleus of ^{212}Po emits α-particles of energy 8.95 MeV and has a half-life of 3.0×10^{-7} seconds. Using the energy-time uncertainty relation

$$\Delta \mathrm{E}\,\tau \geq \hbar/2,$$

   where τ is the average lifetime of the state, calculate the relative spread in energy, ΔE/E, of the α-particles emitted from each of these nuclei.

2. Starting from the classical expression for orbital angular momentum,

$$\mathbf{L} = \mathbf{r} \wedge \mathbf{p}\,,$$

   derive expressions for the quantum mechanical operators $\hat{\mathrm{L}}_x$, $\hat{\mathrm{L}}_y$ and $\hat{\mathrm{L}}_z$. Further show that $\hat{\mathrm{L}}_z$ has the form

$$\hat{\mathrm{L}}_z = -i\hbar \frac{\partial}{\partial \phi}$$

   in a spherical polar coordinate system.

3. Using the expressions for $\hat{\mathrm{L}}_z$ and $\hat{\mathrm{L}}^2$ in the spherical polar coordinate system:

$$\hat{\mathrm{L}}_z = -i\hbar \frac{\partial}{\partial \phi}\,,$$

$$\hat{\mathrm{L}}^2 = -\hbar^2 \left(\frac{\partial^2}{\partial \theta^2} + \cot\theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial \phi^2} \right),$$

   show that the wavefunction

$$\psi(r,\theta,\phi) = R(r)\sqrt{\frac{3}{8\pi}} \sin\theta\, e^{i\phi}$$

   is an eigenfunction of both $\hat{\mathrm{L}}_z$ and $\hat{\mathrm{L}}^2$ with eigenvalues of $+\hbar$ and $2\hbar^2$, respectively.

4. For the wavefunction of question 3, calculate the angle between the z-axis and the direction of the orbital angular momentum vector **L**. Calculate the value of $(L_x^2+L_y^2)$ and comment on the values of the separate components L_x and L_y.

5. Consider a particle moving in a circle with tangential momentum p_t and angular momentum, $L = r\,p_t$. Let Δs be the uncertainty in its position along the circumference and ϕ be the angular position of the particle around the circle.

   (a) Show that the relation $\Delta s\,\Delta p_t \geq \hbar/2$ can be rewritten as $\Delta\phi\,\Delta L \geq \hbar/2$.

   (b) For an electron in the ground state of a hydrogen atom (Bohr orbit), what does this relation imply about the angular position of the electron.

   (c) An electron wavefunction consists of a linear superposition of angular momentum eigenfunctions. The angular position of this electron is known with an uncertainty of $\Delta\phi = (1/6)$ radians. Estimate the range of angular momenta present in the superposition.

6. A beam of hydrogen atoms (spin $\frac{1}{2}\hbar$) is passed through a Stern-Gerlach experiment where the field gradient in the z-direction is $10\,\mathrm{Tm}^{-1}$. The atoms come from an oven at a temperature of 500 K and the magnetic field is 0.5m long. Two images of the beam are found on a screen a distance 2m from the exit of the magnet. Calculate the separation of the images.

   [Bohr magneton = 0.927×10^{-23} amp-m^2
   $g_s = 2.00$
   proton mass = 1.67×10^{-27} kg]